

H1: Custom Program Pitch

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Overview

This task encourages you to showcase your capabilities and growth by designing and building a program of your own invention.

Do the following:

- Start by watching the videos of work from previous students in our [Project Showcase](#).

Tip: Remember this is all about you and your capabilities

While this is all about building a custom program, remember the HD assessment is really all about you showcasing your abilities and your growth.

- Understand the constraints we want you to work within. These include that your program must:
 - Be written in C/C++.
 - Primarily use structured, procedural programming concepts.
 - It can use fixed/dynamic arrays with an object-oriented style, but
 - Must not use object oriented programming in its model.
 - Be functional, though it does not need to be complete. It will most likely be a work in progress, and this is great!
 - Make use of structs to model system entities.
 - Make use of arrays and/or collection classes to work with multiple values.
 - Have its functionality divided across multiple functions and procedures.
 - Have used an iterative development process.
 - Follow good programming practices, including (but not limited to):
 - Following naming and indentation conventions,
 - Not using any global variables,
 - Avoiding duplicating code through appropriate functional abstractions.
 - Be of a size and complexity that demonstrates mastery of these concepts and tools.
 - Be written clearly using things you have learned in this unit:
 - It needs to keep using the convention and style we have used: including limitations on use of the standard library.
 - It must not be a Web Application - even using frameworks - as these are generally not considered suitable for this task.
 - It must not use a game engine (e.g., Unreal/Unity, Godot). Something like SplashKit is okay, but large commercial engines make it too hard to show off what you can do.

Tip: Showing off what you have learnt

We acknowledge that going forward you should make use of the standard library of the language you are using. For this task, however, we want it to be something that looks and feels like an amazing piece of work in FIT1045 - so make use of the things you have used throughout the unit.

- Complete the following short pitch (25 words or less) that highlights the capabilities and growth that you are presenting. Start with your name, and replace the “...” with the details of what you hope to show.

> Alston Liow is awarded “High Distinction” for designing a Gomoku game engine with UI in FIT1045 Introduction to Programming.

- Now elaborate on this - how will you be showcasing these capabilities and growth in the custom program you are planning?

> I will build a complete Gomoku game in C++ using splashkit.
First, I will build a core Gomoku engine which include the logic of the game, separated into three part, board representation, win detection and main logic.
Next, is the UI using splashkit, can handle two player in local.
Finally, I will add an AI opponent using the Minimax algorithm with alpha-beta pruning, enabling player vs computer (PvC) matches.

Tip: Need help and ideas?

Remember you can make use of the teaching team and your peers for ideas and support - though you will need to demonstrate you can execute on this.

- What will the model of your program include?

- **Gomoku Engine:**
 - **Board:** 2D array, store in enum value (empty, black, white)
 - **Placement:** A struct of position and player of single stone placement, can record the history and represent the move.
 - **Placement list:** A dynamic array of placement support undo redo and the minimax algorithm
 - **Game state:** Enum include playing, win, lose
 - **Player:** Enum include black and white
 - **Minimax and alpha-beta pruning**

Reflect:

- What was the most interesting thing you learnt through this task, and why?

> I'm interesting in the chess game, first I consider about Chinese Chess "XiangQi", Chess or the Go, but I think is too complex to do the game logic. So I change to Gomoku so I can spend more time for other advanced features not just on the game engine. I learned about the time planing and how to separate the program into small task.

- What are you most looking forward to in terms of this task?

> I'm looking forward to the Minimax algorithm part. I also want to see how far I can push some machine learning in this project if I have time to do.

- Looking back, how long did it take to complete this task? What did you spend the most time on?

> This planing is using my short free time, to imagine the thing I want and think about the process. I spend most time in writing the actual planing and structure.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I ask Claude about the minimax algorithm.

Tip: Remember we want to see *your* capabilities

Going forward with this project, make sure that you are really controlling how you integrate any use of AI. You will need to convince the teaching team of *your* mastery. In this context, a smaller program that you fully understand is likely to be more convincing than a polished looking program that was largely built by AI.

Remember it is you who gets the HD, not your program!

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack: